Solutions · Series #01

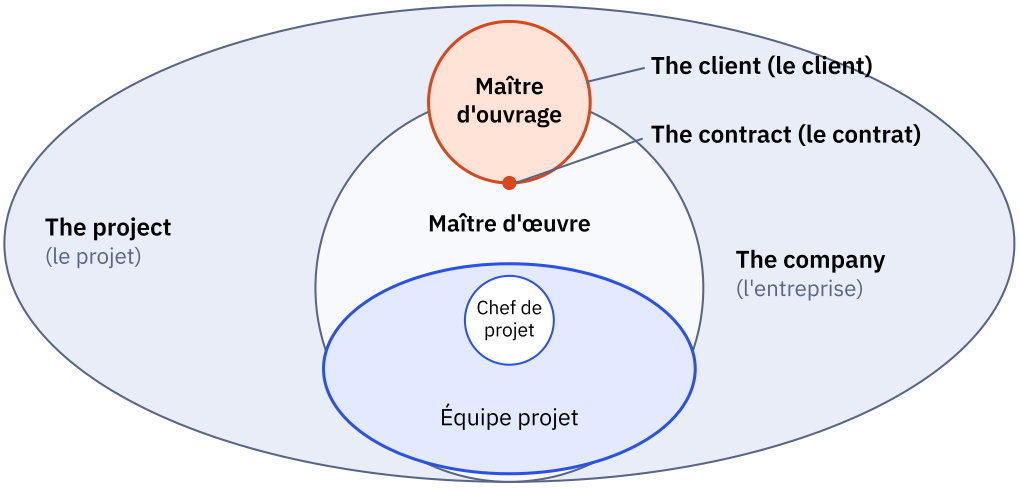
Series #01 · Project management fundamentals and launching *(Série TD #01 · Fondamentaux et lancement de projet)*

Scope. Through these exercises you will discover the main concepts of project fundamentals and of launching a project. *(Concepts fondamentaux et lancement de projet.)*

1.1 · Project stakeholders: MOA and MOE | 1.2 · From an idea to a project | 1.3 · Project or not?

1.1 Project stakeholders: MOA and MOE *(maîtrise d'ouvrage, maîtrise d'œuvre)*

Goal: capture the ecosystem of a project in terms of MOA (project owner, contracting authority) and MOE (prime contractor). Use the stakeholders figure of Chapter 1.



Stakeholders of a project (représentation des parties prenantes d'un projet), redrawn from Chapter 1 of the course.

Q1 Discuss the figure: identify every element and give an example of each.
(Analysez le schéma : identifiez chaque élément et donnez un exemple correspondant.)

SOLUTION

Running example: **building a new library for USTHB**.

Element	Role	Example
The project (le projet)	The whole undertaking: it contains both the client side and the supplier side.	"New USTHB library, 24 months, budget fixed by the ministry"
The client (le client)	Organisation or person that receives the product.	USTHB (and behind it the Ministry of Higher Education)
Maître d'ouvrage (MOA, project owner)	Represents the client inside the project: orders the work, follows progress, attends reviews, controls finances, checks the result meets the needs.	The university's works directorate, led by a project director
The contract (le contrat)	The interface that links client and supplier: scope, cost, deadline, quality.	The public works contract signed after the tender

The company (l'entreprise)	The supplier organisation that provides the product.	A construction company
Maître d'œuvre (MOE, prime contractor)	The part of the company, chosen for its technical skill, that executes the work (or has it done) within time, cost and quality.	The company's engineering and works department
Chef de projet (project manager)	Leads the project team on the supplier side, from A to Z.	The site engineer in charge of the library
Équipe projet (project team)	The people who do the work.	Architects, engineers, foremen, workers

The MOA circle sits inside "the client" and touches the MOE circle: the point where they meet is the contract. The project manager and team sit inside the MOE, which sits inside the company.

Note on the original. Question 2 of this exercise ("continuing the publishing example in question 2, a reviewed article can be rejected, approved, or amendments requested…") refers to an example that does not exist in this series and asks nothing. It seems pasted from another TD, so it is left out here. The original also translates MOE as "Project Manager"; MOE is the *prime contractor* (the supplier side), not the chef de projet.

1.2 From an idea to a project *(de l'idée au projet)*

Goal: see how project ideas emerge and turn into goals after studying different scenarios.

Situation. Ahmed, Riadh and Sara are siblings. Their father is well established in construction and owns a SARL company. He wants to help his children create a business based on their studies, skills and experience, summarised below (father included).

Ahmed, Riadh et Sara sont une fratrie dont le père est bien ancré dans le domaine du bâtiment et possède sa propre SARL. Il voudrait lancer ses enfants dans la création d'un business selon leurs études, compétences et expérience.

Person	Studies *(études)*	Skills *(compétences)*	Experience *(expérience)*
Ahmed	Finance	General accounting *(comptabilité générale)*	2 years
		Financial management *(gestion financière)*	5 years, in his father's company
Riadh	Computer science *(informatique)*	Network development *(développement réseaux)*	7 years, in several companies
		Marketing *(commercialisation)*	3 years*
Sara	Architecture *(architecte)*	Architectural planning *(plan d'architecture)*	2 years
Father	Building *(bâtiment)*	Construction and site supervision *(construction et suivi des travaux)*	more than 20 years; owns a SARL construction company

** The original table also shows a stray "5 years" under Riadh with no skill next to it.*

Q1 From the table, which field or business would suit the project? Why?
(D'après le tableau, quel domaine ou business pourrait convenir au projet ? Pourquoi ?)

```
SOLUTION
```

A building-sector business, for example a real-estate development and design company offering **"smart", connected buildings** (promotion immobilière / bâtiment intelligent).

- **Father**: 20+ years in construction and site supervision, an existing company, a network of suppliers and clients.
- **Sara**: architectural plans, the design part.
- **Riadh**: networks (cabling, home automation, building IT) and marketing (selling the units).
- **Ahmed**: accounting and financial management, already experienced in the father's company.

Together they cover the whole chain: design → build → equip → sell → manage the money. The business is new for the siblings yet builds on the father's experience, which lowers the risk.

Q2 Give possible directions for carrying out this project. The ideas must fit within the targeted business.

(Donnez les orientations possibles pour réaliser ce projet. Les idées doivent s'inscrire dans la sphère du business visé.)

SOLUTION

1. **Real-estate development** (promotion immobilière): design, build and sell or rent housing units.
2. **Architecture and engineering office** (bureau d'études): plans, permits and site supervision for other clients, led by Sara with the father.
3. **Smart buildings / home automation** (domotique): network cabling, security, connected equipment in new and existing buildings, led by Riadh.
4. **Renovation and fit-out** of existing buildings.
5. **Digital sales and property management platform**: online marketing and management of the units (Riadh + Ahmed).

They can start with one direction (e.g. 1 or 2) and add others as phases.

Q3 The father agrees to finance the construction of a building that will be the company's head office. The land is in Ahmed's name. The siblings explain that having premises is the first step of their business, which makes it a sub-project. In the "Head Office Construction" sub-project, who is the MOA, the MOE and the sponsor (mandant)?

(Le père s'engage à financer la construction du siège social ; le terrain est au nom d'Ahmed. Dans le sous-projet « Construction Siège Social », qui sera la MOA, la MOE et le mandant ?)

SOLUTION

Role	Who	Why
MOA (project owner)	The siblings' new company, represented by **Ahmed**	It is the client that orders and receives the building; Ahmed owns the land, so he signs as owner.
MOE (prime contractor)	The **father's SARL**, with Sara for the plans	It has the technical skill to execute the works within time, cost and quality.
Sponsor (mandant)	The **father**	He commits to the sub-project and provides the resources (financing); he answers for its launch.

Accept also: the siblings jointly as sponsor (they asked for the sub-project) with the father as funder. The key point is to keep the client side (MOA) separate from the side that builds (MOE), even within one family, and to formalise it in a contract.

Q4 If the father's funding is not enough, what other means can the siblings use?

(Au cas où le financement du père ne suffirait pas, quels autres moyens la fratrie pourra-t-elle utiliser ?)

SOLUTION

- **Bank loan** (crédit bancaire), using the land or building as a guarantee; also Islamic finance offers.
- **State support schemes for young entrepreneurs**, such as ANADE (formerly ANSEJ), CNAC or ANGEM, and the start-up label with the Algerian Startup Fund for an innovative (smart-building) activity.
- **Leasing** for equipment and vehicles.
- **Partners or investors** who take a share of the company.
- **Phasing**: build in stages, or rent temporary premises and build later.
- **Pre-sales** (vente sur plan) of units once the company starts development.

1.3 Project or not? *(projet ou non-projet)*

Goal: apply the launch criteria to real proposals.

Q1 Briefly recall the criteria that must be known to define a project during its launch phase.
(Rappelez brièvement les critères devant être connus pour définir un projet lors de son lancement.)

SOLUTION

A project exists when all of these are known:

1. it is **unique** (one product or service);
2. it involves some **novelty**;
3. there is a **sponsor** (mandant, commanditaire);
4. there is a **project manager** (chef de projet);
5. a clearly defined, measurable **goal**;
6. a defined **deadline** (délai);
7. a defined **budget**.

Red flags: routine work, no end date, very small amount, vague goal.

Q2 For each proposal in the table, say whether it is a feasible project or not, justify, and propose changes where needed.
(Indiquez parmi ces buts ceux qui peuvent être considérés comme des projets faisables ou pas, en justifiant et en proposant si nécessaire les modifications.)

N°	Proposal	Verdict	Why	Proposed rewording
1	Reduce the primary-school failure rate by 50% in the next 30 years.	**Not as stated**	Measurable, but 30 years is far beyond any project horizon, the scope is national, and there is no sponsor or budget. It is a long-term policy (programme), not a project.	Split into projects, e.g. "Cut the failure rate of 5th-year pupils by 10% in the 40 primary schools of one wilaya within 3 years, with a tutoring programme funded by the Ministry of Education."
2	Clean a facility in at most 1 hour.	**Not a project**	Cleaning is a routine, repetitive task: nothing unique or new, and the amount is tiny.	Keep it as an operational task. A project could be: "Set up a new cleaning procedure and train the staff of the faculty within 2 months."
3	Improve the quality of care in all Algerian hospitals.	**Not a project**	Too vague: "quality of care" is not measured, there is no deadline, no budget, and the scope (all hospitals) is huge.	"Reduce the average emergency waiting time from 3 h to 1.5 h in 3 pilot hospitals of Algiers within 18 months."

4	Reduce the road-accident death rate in the north of the country by about 15% (±) over the next two years.	**Project**	Clear measurable goal (about 15%), defined area (north) and deadline (2 years). Unique and new action.	Make it precise: fix the baseline year and the exact target, name the sponsor (e.g. the national road-safety authority) and the budget. "About ±" should become a firm figure.
5	Increase profits in a manufacturing company.	**Not a project**	No figure, no deadline, no defined scope: a permanent business objective.	"Raise net profit by 10% within 12 months by cutting the scrap rate of production line 2 from 8% to 4%."
6	Increase the scientific visibility of USTHB by 80% in the next five years.	**Project, once reworded**	Measurable target and deadline exist, but "scientific visibility" has no indicator, so 80% means nothing yet.	Define the indicator: "Increase the number of USTHB papers indexed in Scopus by 80% within 5 years, compared with 2025."
7	Switch from teaching in French to teaching in English in the coming years.	**Not as stated**	"In the coming years" is no deadline, and the scope (all teaching? all levels?) and budget are not defined.	"From September 2027, teach all M1 modules of the Faculty of Computer Science in English, after training 60 teachers in 18 months."
8	Build 5 seawater desalination plants in the western region of the country in 36 months.	**Project**	Unique, new, measurable (5 plants), located (west) and dated (36 months).	Only the sponsor (e.g. the Ministry of Water Resources) and the budget remain to be stated.